

Skills

- 6 years experience with SolidWorks and AutoCAD
- Experienced with Microsoft Office
- PLC programming via university projects
- MATLAB, Python, C++ through university projects

Education

MSc in Robotics

Jan 2024 - Jan 2025

National University of Singapore

- Studied Linear Systems, Mathematics for Robotics Engineering, Materials in Robotics
- Designed a decoupled observer-based control system for a chemical reactor using Simulink and MATLAB

BASc in Mechanical Engineering, Mechatronics Option (with Distinction)

Sep 2018 - Jun 2023

University of Waterloo

- Wired, soldered, and programmed Arduino microcontroller based alert system for kelp harvester
- Designed, built, and programmed a LEGO Mindstorms robot with RobotC to autonomously pollinate flowers
- MasterCam via Numerical Control of Machine Tools course
- Ladder logic, structured text and sequential function chart PLC programming

Experience

Mechanical Engineer, Alltech Automotive, Mundelein IL

Aug 2023 - Dec 2023

- Inspected and troubleshoot hundreds of automotive parts for quality control
- Revised and proofread drawings and other technical documents translated from Mandarin to English

Chassis & Dynamics Designer, UWaterloo Baja SAE, Waterloo ON

May 2022 - Aug 2022

- Ensured front suspension system could withstand offroading forces via Finite Element Analysis (FEA) and design iterations in SolidWorks
- Optimized front suspension design for ease of manufacturability

Maintenance Engineering Intern, Owens Corning, Toronto ON

Sep 2021 - Dec 2021

- Saved company over \$10 000 by redesigning winch mounting frame ordered from manufacturer
- Drafted equipment parts in AutoCAD and SolidWorks to communicate needs to external vendors
- Create piping and instrumentation diagram in Microsoft Visio to streamline inter-department communication

Technical Publications Intern, Linamar - Skyjack, Guelph ON

Sep 2019 - Dec 2019

- Took initiative to create VBScript program reducing file processing time from 90 minutes to 5 minutes
- Employed Draftsight and SolidWorks to correct drawings and models, ensuring accuracy in production
- Revised and updated dozens of manuals with Adobe Indesign, Illustrator, and Acrobat to reflect product changes

Cybersecurity Administrator, Enbridge, Chatham ON

Jan 2019 - Apr 2019

- Designed, tested, and conducted phishing simulation for over 10 000 employees
- Researched, created, and presented cybersecurity presentation, developing oral presentation skills
- Improved communication of IT policies using engaging, accessible content